Project Initialization and Planning Phase

Date	27 June 2026
Team ID	
Project Title	Rising Waters
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to develop a flood prediction system using machine learning, improving prediction accuracy and disaster preparedness. It addresses challenges in flood forecasting, enabling timely warnings, reduced risks, and better public safety. Key features include a machine learning-based prediction model and a Flask-based web application for flood risk assessment.

Project Overview	
Objective	To develop a machine learning-based flood prediction system that accurately predicts flood risk using historical weather and rainfall data, enabling timely warnings and effective disaster management.
Scope	The project focuses on analyzing environmental data, training machine learning models, and deploying the best-performing model through a Flask web application for flood prediction.
Problem Statement	
Description	Frequent floods caused by heavy rainfall, climate change, and poor drainage lead to significant loss of life, property, and infrastructure. Existing warning systems may not provide timely and accurate flood risk predictions.
Impact	An accurate flood prediction system helps authorities issue early warnings, reduce disaster risks, protect lives and property, and improve emergency preparedness.
Proposed Solution	
Approach	Use machine learning algorithms to analyze historical rainfall and weather data, train classification models, and deploy the best model using Flask to predict flood risk.
Key Features	• Machine learning-based flood prediction • Historical weather data analysis • Data visualization and preprocessing

	• XGBoost classification model • Flask web application for prediction • Early flood risk assessment

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	6 GPU
Memory	RAM specifications	16 GB
Storage	Disk space for data, models, and logs	1 TB
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	Pandas, numpy,scikit-learn,xgboost,joblib,openpyxl
Development Environment	IDE	VS Code
Data		
Data	Source, size, format	Kaggle Flood Prediction Dataset (.csv/.xlsx)